

Dyadic Calculus for the Fabius Function

Connections among exact values, Appell blocks, Thue–Morse differences, Gaussian interpolation, and discrete limits

A complementary article to *The Fabius Function and Rvachev’s up-Function*

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Abstract

The exact formulae for the values of the Fabius function at dyadic rationals, the binary-reduction series at an arbitrary argument, and the discrete-limit formula involving Gaussian binomial coefficients look at first like three rather different constructions. They are, in fact, organized by one finite polynomial identity and one regular summability argument.

The finite identity is most transparent after introducing the Appell moment polynomials

$$A_n(\theta) = \mathbb{E}(X + \theta)^n,$$

where X has the Fabius distribution. The Taylor-reduction polynomial at scale n is exactly $2^{-\binom{n}{2}} A_n(2^n y)/n!$. Consequently, the global binary expansion becomes a rapidly convergent, digit-sparse Appell series; for a dyadic rational it terminates, so the finite dyadic formula is literally the terminating case of the arbitrary-argument expansion. Replacing each Appell polynomial by its finite Thue–Morse/Gaussian-binomial representation gives the nested q -series term by term, not merely after taking a limit.

The discrete-limit formula uses the same finite Gaussian stencil in a different way. After an exact reindexing it is a signed Toeplitz average of shifted uniform-spline approximants. Its row-generating polynomial is

$$\frac{(z/2; 1/2)_n}{(1/2; 1/2)_n},$$

which simultaneously explains unit row mass, bounded total variation, and the appearance of q -Pochhammer symbols. This article develops these connections, derives quantitative error bounds, and adds several alternative exact representations: a finite composition sum obtained directly from the moment product, a lower-Hessenberg determinant, a Bernoulli–Bell-polynomial formula, and Fourier and Laplace inversion formulae.

Contents

1 Orientation and notation

1.1 What is being connected

The companion article and the Lean development establish, among many other facts, the following three families of formulae.

1. Exact rational expressions for $F(m/2^n)$, including formulae in which a Gaussian binomial coefficient at base $1/2$ multiplies a finite signed Thue–Morse power sum.
2. An exact infinite expansion for $F(x)$, or for the signed global extension $\mathcal{F}(x)$, obtained by repeatedly reducing along the binary expansion of x . Substituting the finite q -binomial expression into every reduction block produces a nested infinite q -series.

3. A discrete limit in which a finite q -binomial sum of increasingly fine Thue–Morse splines converges to $\mathcal{F}(x)$ at an arbitrary real argument.

The purpose here is not to repeat the proofs already presented there. We instead isolate the common algebraic mechanism, compress the formulae into a small collection of reusable objects, and then unfold those objects in several new directions.

Throughout,

$$\rho = \frac{1}{2}, \quad (a; \rho)_n = \prod_{j=0}^{n-1} (1 - a\rho^j), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_{\rho} = \frac{(\rho; \rho)_n}{(\rho; \rho)_k (\rho; \rho)_{n-k}}.$$

The empty product is 1. We write

$$\varepsilon_r = (-1)^{\text{wt}_2(r)},$$

where $\text{wt}_2(r)$ is the number of 1's in the binary expansion of r . Thus $(\varepsilon_r)_{r \geq 0}$ is the $\{\pm 1\}$ Thue–Morse sequence.

1.2 The half-base notation is Mersenne arithmetic

At the special base $\rho = 1/2$, the q -Pochhammer symbol is simply a compact normalization of a product of Mersenne numbers:

$$(1/2; 1/2)_n = \prod_{j=1}^n (1 - 2^{-j}) = 2^{-\binom{n+1}{2}} \prod_{j=1}^n (2^j - 1). \quad (1)$$

In particular,

$$2^{n^2} (1/2; 1/2)_n = 2^{\binom{n}{2}} \prod_{j=1}^n (2^j - 1). \quad (2)$$

The factors $2^j - 1$ appearing separately in moment recurrences and composition formulae are therefore the same arithmetic content that the q -Pochhammer symbol packages in the closed Gaussian-binomial formula. Likewise,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2} = 2^{-k(n-k)} \left[\begin{matrix} n \\ k \end{matrix} \right]_2. \quad (3)$$

Thus the half-base Gaussian coefficients are ordinary integral Gaussian coefficients at base 2, accompanied by an explicit power of 2. The q -notation is not introducing mysterious irrational constants; it records the geometry of the dyadic interpolation grid and keeps its Mersenne normalizations factored.

1.3 The probabilistic normalization

Let $U_1, U_2, \dots$ be independent random variables, each uniformly distributed on $[0, 1]$, and put

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}. \quad (4)$$

Then $0 \leq X \leq 1$, and the distribution function of X is the Fabius function:

$$F(x) = \mathbb{P}(X \leq x).$$

This representation fixes all normalizations used below. Define the moments

$$d_n = \mathbb{E}X^n, \quad d_0 = 1, \quad (5)$$

and their exponential generating function

$$G(z) = \mathbb{E}e^{zX} = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!}. \quad (6)$$

Independence in (??) gives the entire product

$$G(z) = \prod_{j=1}^{\infty} E\left(\frac{z}{2^j}\right), \quad E(w) = \frac{e^w - 1}{w}. \quad (7)$$

Shifting the product by one factor yields

$$G(2z) = E(z)G(z). \quad (8)$$

Comparing coefficients gives the basic moment recurrence

$$(2^n - 1) \frac{d_n}{n!} = \sum_{k=0}^{n-1} \frac{d_k}{k!(n-k+1)!} \quad (n \geq 1). \quad (9)$$

One of the fundamental exact dyadic identities is

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{d_n}{n!}. \quad (10)$$

It will reappear in nearly every section, but in different disguises.

The product (??), the recurrence (??), the dyadic identity (??), the finite q -binomial formulae, and the spline limits are all formally proved in the cited Lean development. Some of the regroupings below—notably the Appell compression, determinant form, and explicit error estimate for the Toeplitz limit—are elementary deductions from those formalized identities and are presented here in human-readable form.

2 The master Taylor block

2.1 The reduction polynomial

For $n \geq 0$, define

$$R_n(y) = \sum_{k=0}^n 2^{\binom{k+1}{2} - \binom{n-k}{2}} \frac{d_{n-k}}{(n-k)!} \frac{y^k}{k!}. \quad (11)$$

This is the polynomial that occurs in the exact Taylor reduction. On the appropriate dyadic shell, if $y = x - 2^{-n}$, the global Fabius extension satisfies an identity of the form

$$\mathcal{F}(x) = -\mathcal{F}(y) + R_n(y).$$

Only the polynomial itself will be needed here.

Putting $y = 0$ in (??) gives

$$R_n(0) = 2^{-\binom{n}{2}} \frac{d_n}{n!} = F(2^{-n}),$$

so every inverse-dyadic value is the constant term of one reduction block. The less obvious fact is that the whole block has an equally simple meaning.

2.2 Appell compression

Definition 2.1 (Moment Appell polynomials). For $n \geq 0$ and $\theta \in \mathbb{C}$, let

$$A_n(\theta) = \mathbb{E}(X + \theta)^n = \sum_{k=0}^n \binom{n}{k} d_{n-k} \theta^k. \quad (12)$$

Equivalently,

$$\sum_{n=0}^{\infty} A_n(\theta) \frac{z^n}{n!} = e^{\theta z} G(z). \quad (13)$$

The family is Appell in the classical sense:

$$A'_n(\theta) = n A_{n-1}(\theta), \quad A_n(\theta + c) = \sum_{j=0}^n \binom{n}{j} A_{n-j}(\theta) c^j. \quad (14)$$

Since X and $1 - X$ have the same distribution,

$$A_n(\theta) = (-1)^n A_n(-1 - \theta). \quad (15)$$

Theorem 2.2 (Appell compression of the reduction block). For every $n \geq 0$ and $\theta \in \mathbb{C}$,

$$R_n(2^{-n}\theta) = 2^{-\binom{n}{2}} \frac{A_n(\theta)}{n!}. \quad (16)$$

Equivalently, for arbitrary y ,

$$R_n(y) = 2^{-\binom{n}{2}} \frac{A_n(2^n y)}{n!}. \quad (17)$$

Proof. Substitute $y = 2^{-n}\theta$ into (??). The power of 2 in the k th term becomes

$$\binom{k+1}{2} - \binom{n-k}{2} - nk = -\binom{n}{2},$$

independently of k . Factoring out $2^{-\binom{n}{2}}$ leaves

$$\sum_{k=0}^n \frac{d_{n-k} \theta^k}{(n-k)!k!} = \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} d_{n-k} \theta^k = \frac{A_n(\theta)}{n!}.$$

□

This elementary cancellation is the main compression used throughout the article. A triangular-looking polynomial with apparently irregular powers of 2 is simply a rescaled shifted moment.

2.3 Immediate consequences

For $0 \leq \theta < 1$, positivity and bounded support give

$$0 \leq A_n(\theta) \leq 2^n. \quad (18)$$

Consequently,

$$0 \leq R_n(2^{-n}\theta) \leq \frac{2^{n-\binom{n}{2}}}{n!}. \quad (19)$$

This estimate is much sharper than the geometric estimate normally used to prove convergence of the binary-reduction series.

The first few Appell polynomials are

$$\begin{aligned} A_0(\theta) &= 1, \\ A_1(\theta) &= \theta + \frac{1}{2}, \\ A_2(\theta) &= \theta^2 + \theta + \frac{5}{18}, \\ A_3(\theta) &= \theta^3 + \frac{3}{2}\theta^2 + \frac{5}{6}\theta + \frac{1}{6}, \\ A_4(\theta) &= \theta^4 + 2\theta^3 + \frac{5}{3}\theta^2 + \frac{2}{3}\theta + \frac{143}{1350}. \end{aligned}$$

They encode all Taylor blocks of the corresponding orders by a single change of scale.

3 The binary expansion as a sparse Appell series

3.1 Prefixes, bits, and tails

Fix $x \in [0, 1)$. Use the canonical binary expansion that is not eventually all 1's,

$$x = 0.b_1b_2b_3 \dots 2, \quad b_m \in \{0, 1\}. \quad (20)$$

Let

$$a_m = \lfloor 2^m x \rfloor, \quad \theta_m = \{2^m x\}, \quad y_m = x - \frac{a_m}{2^m} = 2^{-m}\theta_m. \quad (21)$$

Then

$$a_m = 2a_{m-1} + b_m, \quad \theta_m = 0.b_{m+1}b_{m+2} \dots 2. \quad (22)$$

The binary-reduction series in the formal development has coefficient

$$c_m(x) = \varepsilon_{a_m}(2a_{m-1} - a_m) \quad (m \geq 1). \quad (23)$$

Using (??), this simplifies completely:

$$c_m(x) = \begin{cases} 0, & b_m = 0, \\ \varepsilon_{a_{m-1}}, & b_m = 1. \end{cases} \quad (24)$$

Indeed, when $b_m = 1$ one has $\varepsilon_{a_m} = -\varepsilon_{a_{m-1}}$ and $2a_{m-1} - a_m = -1$. Thus only the positions of the 1-bits contribute.

3.2 The digit–Appell formula

Theorem 3.1 (Digit-sparse Appell expansion). *For every $x \in [0, 1)$,*

$$F(x) = \sum_{m=1}^{\infty} b_m (-1)^{b_1+\dots+b_{m-1}} \frac{2^{-\binom{m}{2}}}{m!} A_m(\theta_m). \quad (25)$$

The series converges absolutely, and in fact uniformly with respect to $x \in [0, 1)$.

Proof. The exact binary-reduction series is

$$F(x) = \sum_{m \geq 1} c_m(x) R_m(y_m).$$

Insert (??), use $\varepsilon_{a_{m-1}} = (-1)^{b_1+\dots+b_{m-1}}$, and then apply ?? with $y_m = 2^{-m}\theta_m$. Absolute and uniform convergence follow from (??) and the Weierstrass test. $\square$

Formula (??) is a compact arbitrary-argument representation that contains no q -notation at all. The q -series will emerge by replacing each A_m by an exactly equal finite Gaussian-binomial expression.

3.3 A superexponential remainder bound

Let $S_N(x)$ denote the truncation of (??) after $m = N$. From (??),

$$|F(x) - S_N(x)| \leq \sum_{m=N+1}^{\infty} \frac{2^{m-\binom{m}{2}}}{m!}.$$

The ratio of consecutive majorants is

$$\frac{2^{m+1-\binom{m+1}{2}}/(m+1)!}{2^{m-\binom{m}{2}}/m!} = \frac{2^{1-m}}{m+1} \leq \frac{1}{2} \quad (m \geq 1).$$

Therefore, for $N \geq 0$,

$$|F(x) - S_N(x)| \leq 2 \frac{2^{N+1-\binom{N+1}{2}}}{(N+1)!}. \quad (26)$$

The right-hand side decays faster than geometrically in N . In practical computation, the binary digit series is therefore far more rapidly convergent than the elementary residual estimate $O(2^{-N})$ suggests.

3.4 Dyadic rationals are the terminating case

Suppose

$$x = \frac{M}{2^N} = 0.b_1b_2 \dots b_N 000 \dots_2, \quad 0 \leq M < 2^N.$$

Then $b_m = 0$ for $m > N$, and (??) terminates:

$$F\left(\frac{M}{2^N}\right) = \sum_{\substack{1 \leq m \leq N \\ b_m=1}} (-1)^{b_1+\dots+b_{m-1}} \frac{2^{-\binom{m}{2}}}{m!} A_m(0.b_{m+1} \dots b_N). \quad (27)$$

Thus the exact bit-recursive evaluator, the finite exact dyadic formula, and the arbitrary-argument series are not separate phenomena: they are the finite, recursive, and infinite readings of the same identity.

Example 3.2 (The value at $5/16$). The terminating binary expansion is

$$\frac{5}{16} = 0.0101_2.$$

Only $m = 2$ and $m = 4$ contribute. At $m = 2$ the previous prefix has even weight and $\theta_2 = 1/4$; at $m = 4$ the previous prefix is 010_2 , so the sign is negative and $\theta_4 = 0$. Hence

$$\begin{aligned} F\left(\frac{5}{16}\right) &= \frac{2^{-1}}{2!} A_2\left(\frac{1}{4}\right) - \frac{2^{-6}}{4!} A_4(0) \\ &= \frac{1}{4} \left(\frac{5}{18} + \frac{1}{4} + \frac{1}{16} \right) - \frac{1}{1536} \frac{143}{1350} \\ &= \frac{305857}{2073600}. \end{aligned}$$

The first contribution accounts for the prefix 01 together with the remaining tail 01 ; the second contribution corrects for the final 1 -bit.

4 Why Thue–Morse and Gaussian binomial coefficients appear

Two independent annihilation mechanisms are superposed in the nested exact formula. The inner Thue–Morse sum is an additive finite-difference operator; the outer Gaussian-binomial sum is a divided-difference operator on a geometric grid. Each kills lower-degree terms. Their composition isolates precisely the coefficient needed by the Taylor block.

4.1 The Thue–Morse difference operator

For a polynomial or entire function P , define

$$(\mathfrak{T}_k P)(c) = \sum_{r=0}^{2^k-1} \varepsilon_r P(c+r). \quad (28)$$

Writing $D = \frac{d}{dc}$ and using the binary digits of r gives the operator factorization

$$\mathfrak{T}_k = \prod_{j=0}^{k-1} (1 - e^{2^j D}). \quad (29)$$

Indeed,

$$\sum_{r < 2^k} \varepsilon_r e^{rz} = \prod_{j=0}^{k-1} (1 - e^{2^j z}).$$

Since every factor in (29) begins with $-2^j D$, the operator annihilates all polynomials of degree less than k and satisfies

$$\sum_{r=0}^{2^k-1} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}} k!. \quad (30)$$

More generally, for fixed n ,

$$c \mapsto \frac{1}{(n+k)!} \sum_{r < 2^k} \varepsilon_r (r+c)^{n+k} \quad (31)$$

has degree at most n , and its leading coefficient is

$$[c^n] \left(\frac{1}{(n+k)!} \sum_{r < 2^k} \varepsilon_r (r+c)^{n+k} \right) = \frac{(-1)^k 2^{\binom{k}{2}}}{n!}. \quad (32)$$

A symmetric form of the operator is also useful. With $\operatorname{sinhc}(z) = \sinh(z)/z$,

$$\mathfrak{T}_k = (-1)^k 2^{\binom{k}{2}} D^k e^{(2^k-1)D/2} \prod_{j=0}^{k-1} \operatorname{sinhc}(2^{j-1} D). \quad (33)$$

This shows that the Thue–Morse block is a k th derivative, followed by an even differential correction and evaluation about the midpoint of the block. It also explains why centered and translated versions of the formula are so natural.

4.2 Geometric divided differences

Let $0 < q < 1$ for a moment, and consider the geometric nodes

$$1, q^{-1}, q^{-2}, \dots, q^{-n}.$$

The leading coefficient of any polynomial P of degree at most n is its n th divided difference on these nodes. Lagrange interpolation gives

$$[t^n] P(t) = \frac{(-1)^n q^{\binom{n+1}{2}}}{(q; q)_n} \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q P(q^{-k}). \quad (34)$$

For completeness, the barycentric denominator is

$$\frac{1}{\prod_{j \neq k} (q^{-k} - q^{-j})} = \frac{(-1)^{n-k} q^{\binom{n+1}{2} + \binom{k}{2}}}{(q; q)_n} \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (35)$$

At $q = 1/2$, the nodes are $1, 2, 4, \dots, 2^n$ and the interpolation weights are exactly Gaussian binomial coefficients at base $1/2$, decorated by triangular powers of 2.

The same identity follows immediately from the finite q -binomial theorem

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k = (z; q)_n. \quad (36)$$

If $z = q^{-m}$, the right-hand side is zero for $m < n$, while

$$(q^{-n}; q)_n = (-1)^n q^{-\binom{n+1}{2}} (q; q)_n. \quad (37)$$

Thus the stencil kills every lower monomial and normalizes the top one.

4.3 The two annihilators together

The inner sum (??) first lowers degree by k . Its remaining dependence is a polynomial of degree at most n . The outer geometric stencil (??) then kills all but one prescribed coefficient of that degree- n polynomial. Schematically,

$\text{power of degree } n + k \xrightarrow{\mathfrak{T}_k} \text{polynomial of degree } n \xrightarrow{\Omega_n} \text{one coefficient.}$

This is the structural reason that a formula about a smooth distribution function involves both Thue–Morse signs and Gaussian binomial coefficients. Neither ingredient is ornamental: each is the coefficient vector of a finite annihilator.

Translation invariance is now transparent. Translating the argument of the intermediate degree- n polynomial changes only lower coefficients after the target coefficient is fixed. The geometric extractor annihilates precisely those changes. The Lean proof strengthens this observation by treating the translation as an indeterminate and proving that the complete numerator polynomial is constant.

5 The finite q -formula as an Appell identity

5.1 A centered master formula

The exact inverse-dyadic formula is most revealing when one retains the Taylor variable instead of setting it to zero.

Theorem 5.1 (Centered Gaussian-binomial representation of A_n). *For every $n \geq 0$ and $\theta \in \mathbb{C}$,*

$$\frac{A_n(\theta)}{n!} = \frac{2^{-(n+1)}}{(1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k(1+\theta))^{n+k}. \quad (38)$$

Proof. The polynomial version of the formalized q -binomial Taylor identity is

$$R_n(y) = \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r < 2^k} \varepsilon_r (r - 2^k - 2^{n+k} y)^{n+k}.$$

Set $y = 2^{-n}\theta$ and use ???. Multiplying by $2^{\binom{n}{2}}$ changes the outer power 2^{-n^2} into $2^{-\binom{n+1}{2}}$, while the inner shift becomes $-2^k(1 + \theta)$. $\square$

At $\theta = 0$, multiplying (??) by $2^{-\binom{n}{2}}$ gives the exact centered inverse-dyadic formula

$$F(2^{-n}) = \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k)^{n+k}. \quad (39)$$

Thus the formula for one special value is merely the constant-term specialization of a formula for the entire Appell polynomial.

5.2 The arbitrary-translation form

The companion development proves an even stronger polynomial identity. A common translation parameter may be inserted into all the raw Thue–Morse blocks without changing the result.

Theorem 5.2 (Raw translated representation). *For every $n \geq 0$ and every $\theta, \eta \in \mathbb{C}$,*

$$\frac{A_n(\theta)}{n!} = \frac{(-1)^n 2^{-\binom{n+1}{2}}}{(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r + \eta + 2^k \theta)^{n+k}. \quad (40)$$

The right-hand side is independent of η .

Putting $\theta = 0$ gives

$$F(2^{-n}) = \frac{1}{(-2)^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r + \eta)^{n+k}. \quad (41)$$

The identity $(-1)^n = (-1)^{n^2}$ accounts for the denominator $(-2)^{n^2}$.

Remark 5.3 (A scale-dependent translation). Because (??) is an identity separately for each n , one may choose a different common translation η_n at every outer scale in an infinite expansion. This does not change a single term. In exact arithmetic this is merely a gauge freedom; in floating-point arithmetic it may be used to reduce the size of intermediate powers. The translation must remain common to the inner k -sum for a fixed n .

6 One-shot and digitwise formulae for a general dyadic rational

The formula (??) treats the numerator 1. There is also a one-shot formula for a general numerator. For $0 \leq M < 2^N$,

$$F\left(\frac{M}{2^N}\right) = \frac{1}{2^{N^2} (1/2; 1/2)_N} \sum_{k=0}^N \frac{[N]_{1/2}}{4^{\binom{k}{2}} (N+k)!} \sum_{r=0}^{M2^k-1} \varepsilon_r (r - M2^k)^{N+k}. \quad (42)$$

A common translation may be inserted in all the inner powers; the complete outer numerator is unchanged. Formula (??) uses the denominator exponent N once and for all. In contrast, (??) uses

one lower-order block for each 1-bit of M . The two organizations are related by the canonical dyadic decomposition of the prefix interval.

Write

$$\frac{M}{2^N} = 0.b_1 \dots b_{N2}, \quad a_m = \lfloor 2^m M / 2^N \rfloor.$$

For every function P on the nonnegative integers and every $k \geq 0$,

$$\sum_{r=0}^{M2^k-1} \varepsilon_r P(r) = \sum_{\substack{1 \leq m \leq N \\ b_m=1}} \varepsilon_{a_{m-1}} \sum_{s=0}^{2^{N-m+k}-1} \varepsilon_s P(a_{m-1} 2^{N-m+k+1} + s). \quad (43)$$

Indeed, the interval $[0, M)$ is the disjoint union, over its 1-bits, of

$$[a_{m-1} 2^{N-m+1}, a_{m-1} 2^{N-m+1} + 2^{N-m}), \quad b_m = 1.$$

After multiplication by 2^k , every block starts at a multiple of twice its length. Hence adding the local coordinate s creates no binary carry and

$$\varepsilon_{a_{m-1} 2^{N-m+k+1} + s} = \varepsilon_{a_{m-1}} \varepsilon_s.$$

This proves (??).

Substituting (??) into the one-shot formula splits every long Thue–Morse prefix into complete blocks carrying exactly the signs $\varepsilon_{a_{m-1}}$ that occur in the digit formula. The remaining outer Gaussian stencil is still of order N ; applying the polynomial Taylor identity to each rescaled block lowers it to its natural bit scale m and produces (??). Thus:

- (??) groups the computation by polynomial degree $N + k$;
- (??) groups it by the occupied binary scales of the numerator;
- (??) is the combinatorial bridge between the two groupings.

For $M = 5$ and $N = 4$, the one-shot expression has one outer sum of order 4 and inner prefixes of lengths $5 \cdot 2^k$. The digitwise expression has only two active blocks, at scales $m = 2$ and $m = 4$, as shown in ??. Their equality is a nontrivial finite rearrangement, not a limiting assertion.

7 From exact dyadic values to the arbitrary-argument q -series

The connection can now be made without any further analysis: substitute ?? or ?? into the digit–Appell series term by term.

7.1 The centered nested series

Theorem 7.1 (Centered digit– q expansion). *For $x \in [0, 1)$, with notation (??)–(??),*

$$\begin{aligned} F(x) = \sum_{m=1}^{\infty} b_m \varepsilon_{a_{m-1}} \frac{1}{2^{m^2} (1/2; 1/2)_m} \sum_{k=0}^m \frac{\lfloor m \rfloor_{1/2}}{4^{\binom{k}{2}} (m+k)!} \\ \times \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k(1 + \theta_m))^{m+k}. \end{aligned} \quad (44)$$

The equality holds term by term with (??).

Proof. In the m th term of (??), multiply the prefactor $2^{-\binom{m}{2}}$ by the prefactor $2^{-\binom{m+1}{2}}$ in (??). Their product is 2^{-m^2} . Everything else is a direct substitution. $\square$

If $x = M/2^N$ has a terminating binary expansion, then (??) terminates after $m = N$. The “infinite” global q -series is therefore exactly a finite dyadic formula whenever its argument is dyadic. Conversely, allowing infinitely many nonzero binary digits turns the same finite block identity into the arbitrary-argument series.

7.2 The raw translated nested series

For any fixed $\eta \in \mathbb{C}$, the raw form is

$$F(x) = \sum_{m=1}^{\infty} b_m \varepsilon_{a_{m-1}} \frac{1}{(-2)^{m^2} (1/2; 1/2)_m} \sum_{k=0}^m \frac{[m]_{1/2}}{4^{\binom{k}{2}} (m+k)!} \times \sum_{r=0}^{2^k-1} \varepsilon_r (r + \eta + 2^k \theta_m)^{m+k}. \quad (45)$$

Again, every individual m th summand equals the corresponding Appell summand in (??); no interchange of infinite sums is needed to prove the identity.

7.3 A useful commutative diagram

The relations may be summarized as follows:

$$\begin{array}{ccc} R_m(2^{-m}\theta) & \xleftarrow{\text{Appell compression}} & 2^{-\binom{m}{2}} A_m(\theta)/m! \\ \uparrow \text{finite Gaussian/Thue–Morse identity} & & \downarrow \text{insert binary tail } \theta=\theta_m \\ \text{finite nested } q\text{-block} & \xrightarrow{\text{sum over the 1-bits}} & F(x). \end{array}$$

Setting $\theta = 0$ in the top row gives $F(2^{-m})$. Taking a terminating binary expansion makes the right-hand summation finite. Taking an arbitrary binary expansion leaves an absolutely convergent infinite series. These are specializations, not analogies.

8 Alternative exact representations of inverse-dyadic values

The q -binomial formula is only one exact representation of $F(2^{-n})$. The moment product (??) leads to several others, each exposing a different combinatorial structure.

8.1 Coefficient extraction from the moment product

Let

$$a_n = \frac{d_n}{n!} = 2^{\binom{n}{2}} F(2^{-n}). \quad (46)$$

Then

$$a_n = [z^n] \prod_{j=1}^{\infty} \left(\sum_{r=0}^{\infty} \frac{z^r}{2^{jr} (r+1)!} \right). \quad (47)$$

Expanding by the set of scales at which a nonconstant term is chosen gives

$$a_n = \sum_{\ell=1}^n \sum_{\substack{r_1+\dots+r_\ell=n \\ r_i \geq 1}} \frac{1}{\prod_{i=1}^{\ell} (r_i+1)!} \sum_{1 \leq j_1 < \dots < j_\ell} 2^{-\sum_{i=1}^{\ell} j_i r_i}, \quad (n \geq 1). \quad (48)$$

The infinite inner sum is an elementary geometric simplex sum. If $t_h = r_h + \dots + r_\ell$, then

$$\sum_{1 \leq j_1 < \dots < j_\ell} 2^{-\sum_i j_i r_i} = \prod_{h=1}^{\ell} \frac{1}{2^{t_h} - 1}. \quad (49)$$

To see this, put $j_i = i + g_1 + \cdots + g_i$ with $g_h \geq 0$; the resulting independent geometric series contribute $(1 - 2^{-t_h})^{-1}$, and their powers of 2 cancel the initial offset.

Reversing the composition $(r_1, \dots, r_\ell)$ changes tail sums into prefix sums. We obtain the finite composition formula

Theorem 8.1 (Composition representation). *Let $\rho = (r_1, \dots, r_\ell) \models n$ run over all ordered compositions of n , and let $s_j = r_1 + \cdots + r_j$. Then*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \models n} \prod_{j=1}^{\ell} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \quad (50)$$

This derivation explains the factors $2^{s_j} - 1$: they are the closed forms of geometric sums over the infinitely many binary scales in (??).

8.2 A lower-Hessenberg determinant

The recurrence (??) may be written

$$a_n = \sum_{k=0}^{n-1} \beta_{n,k} a_k, \quad \beta_{n,k} = \frac{1}{(2^n - 1)(n - k + 1)!}. \quad (51)$$

Define the $n \times n$ lower-Hessenberg matrix

$$H_n = \begin{pmatrix} \beta_{1,0} & -1 & 0 & \cdots & 0 \\ \beta_{2,0} & \beta_{2,1} & -1 & \ddots & \vdots \\ \beta_{3,0} & \beta_{3,1} & \beta_{3,2} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ \beta_{n,0} & \beta_{n,1} & \beta_{n,2} & \cdots & \beta_{n,n-1} \end{pmatrix}. \quad (52)$$

Proposition 8.2 (Determinant representation). *For $n \geq 1$,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \det H_n. \quad (53)$$

Proof. Let $D_0 = 1$ and $D_n = \det H_n$. Expanding the determinant along its last row, with the -1 superdiagonal supplying the compensating cofactor signs, gives

$$D_n = \sum_{k=0}^{n-1} \beta_{n,k} D_k.$$

This is exactly (??), with the same initial value; hence $D_n = a_n$. $\square$

Expanding the determinant by paths through the nonzero Hessenberg band recovers the composition sum (??). Thus the composition and determinant formulae are two standard expansions of the same triangular resolvent.

8.3 Cumulants, Bernoulli numbers, and Bell polynomials

Taking the logarithm of (??) gives another exact description. The classical expansion

$$\log \frac{e^z - 1}{z} = \frac{z}{2} + \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!} z^{2m} \quad (54)$$

uses the Bernoulli numbers B_{2m} . Summing over the dyadic scales yields

$$\log G(z) = \frac{z}{2} + \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2^{2m} - 1)} \frac{z^{2m}}{(2m)!}. \quad (55)$$

Therefore the cumulants of X are

$$\kappa_1 = \frac{1}{2}, \quad \kappa_{2m} = \frac{B_{2m}}{2m(2^{2m} - 1)}, \quad \kappa_{2m+1} = 0 \quad (m \geq 1). \quad (56)$$

The vanishing odd cumulants beyond the first are another expression of the symmetry $X \stackrel{d}{=} 1 - X$.

Let $\mathcal{B}_n$ denote the complete exponential Bell polynomial, normalized by

$$\exp \left(\sum_{j=1}^{\infty} x_j \frac{z^j}{j!} \right) = \sum_{n=0}^{\infty} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (57)$$

Then $d_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n)$, and hence

Theorem 8.3 (Bernoulli–Bell representation).

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad (58)$$

where the κ_j are given by (??).

Equivalently, expanding the complete Bell polynomial gives the finite partition sum

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\substack{\nu_1, \dots, \nu_n \geq 0 \\ \nu_1 + 2\nu_2 + \dots + n\nu_n = n}} \prod_{j=1}^n \frac{1}{\nu_j!} \left(\frac{\kappa_j}{j!} \right)^{\nu_j}. \quad (59)$$

Only κ_1 and the even cumulants are nonzero, so the apparent n -variable sum is substantially sparse.

For example,

$$d_2 = \kappa_1^2 + \kappa_2 = \frac{5}{18}, \quad d_4 = \kappa_1^4 + 6\kappa_1^2\kappa_2 + 3\kappa_2^2 + \kappa_4 = \frac{143}{1350}.$$

Consequently,

$$F\left(\frac{1}{4}\right) = \frac{5}{72}, \quad F\left(\frac{1}{16}\right) = \frac{143}{2073600}.$$

The Bell-polynomial formula replaces Thue–Morse cancellation by cumulant addition: the dyadic scales contribute independently to $\log G$.

9 The discrete limit as a q -Toeplitz transform

The discrete limit uses the same finite Gaussian coefficients as the exact formula, but its inner sums are truncated at an x -dependent point rather than taken over a complete Thue–Morse block. Exact coefficient extraction is therefore replaced by convergence of splines and regular summability.

9.1 Shifted finite splines

For $p \geq 0$, $\eta \in \mathbb{C}$, and real x , define

$$S_{p,\eta}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{0 \leq r < \lfloor 2^p x + 1/2 \rfloor} \varepsilon_r (r - 2^p x + \eta)^p. \quad (60)$$

An empty sum is zero. The centered choice is $\eta = 1/2$. The formal development proves the uniform estimate

$$|S_{p,1/2}(x) - \mathcal{F}(x)| \leq 2^{-p} \quad (x \in \mathbb{R}), \quad (61)$$

and, for fixed complex shift and $p \geq 1$,

$$|S_{p,\eta}(x) - S_{p,1/2}(x)| \leq 2^{1-p} e^{|\eta-1/2|} \quad (x \geq 0). \quad (62)$$

On the negative half-line the relevant sums and the global extension vanish, so the convergence statement extends without difficulty.

9.2 The discrete q -sum

Define

$$\begin{aligned} D_n(\eta, x) &= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ &\quad \times \sum_{0 \leq r < \lfloor 2^{n+k} x + 1/2 \rfloor} \varepsilon_r (r - 2^{n+k} x + \eta)^{n+k}. \end{aligned} \quad (63)$$

This looks like a direct arbitrary- x analogue of (?). The decisive simplification is an exact reindexing.

Theorem 9.1 (Exact Toeplitz decomposition). *Put*

$$w_{n,j} = \frac{(-1)^j [n]_{1/2} 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n} \quad (0 \leq j \leq n). \quad (64)$$

Then

$$D_n(\eta, x) = \sum_{j=0}^n w_{n,j} S_{2n-j,\eta}(x). \quad (65)$$

Proof. Set $j = n - k$ and $p = n + k = 2n - j$ in (?). Replacing the inner sum by the definition of $S_{p,\eta}$ leaves the power of 2

$$\binom{2n-j}{2} - n^2 - 2 \binom{n-j}{2} = -\binom{j+1}{2},$$

and the sign $(-1)^{2n-j} = (-1)^j$. Symmetry of the Gaussian binomial coefficient gives $[n]_{1/2} = [n-j]_{1/2}$. $\square$

The indices $2n - j$ lie between n and $2n$. Thus every spline entering the n th row is already at resolution at least n .

9.3 One polynomial explains the entire row

Define the row-generating polynomial

$$W_n(z) = \sum_{j=0}^n w_{n,j} z^j. \quad (66)$$

The finite q -binomial theorem gives the compact product

Theorem 9.2 (Toeplitz row polynomial).

$$W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}. \quad (67)$$

Three important facts are immediate.

First, the row mass is one:

$$\sum_{j=0}^n w_{n,j} = W_n(1) = 1. \quad (68)$$

Second, the signs alternate, so the exact total variation is

$$V_n := \sum_{j=0}^n |w_{n,j}| = W_n(-1) = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n}. \quad (69)$$

The sequence V_n is bounded; the formal proof uses the convenient uniform bound $V_n \leq 16$.

Third, differentiating at $z = 1$ gives the first signed moment of the row:

$$\sum_{j=0}^n j w_{n,j} = W'_n(1) = - \sum_{\ell=1}^n \frac{1}{2^\ell - 1}. \quad (70)$$

Higher factorial moments follow by further logarithmic differentiation of (??). Thus the entire Toeplitz stencil is encoded by one finite q -Pochhammer ratio.

The weights also admit the symmetric form

$$w_{n,j} = \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_j (1/2; 1/2)_{n-j}}. \quad (71)$$

For fixed j they converge to

$$w_{\infty,j} = \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_j (1/2; 1/2)_{\infty}}, \quad (72)$$

whose generating function is

$$W_{\infty}(z) = \frac{(z/2; 1/2)_{\infty}}{(1/2; 1/2)_{\infty}}. \quad (73)$$

This limiting object is a stationary signed q -filter of unit mass and finite variation.

9.4 A quantitative discrete-limit estimate

The exact Toeplitz decomposition makes an explicit error bound immediate.

Theorem 9.3 (Uniform rate for fixed shift). *For $n \geq 1$, $x \geq 0$, and $\eta \in \mathbb{C}$,*

$$|D_n(\eta, x) - \mathcal{F}(x)| \leq V_n \left(2^{-n} + 2^{1-n} e^{|\eta-1/2|} \right), \quad (74)$$

where V_n is given by (??). In particular,

$$|D_n(\eta, x) - \mathcal{F}(x)| \leq 16 \left(2^{-n} + 2^{1-n} e^{|\eta-1/2|} \right). \quad (75)$$

Proof. By (??) and ??,

$$D_n(\eta, x) - \mathcal{F}(x) = \sum_{j=0}^n w_{n,j} (S_{2n-j,\eta}(x) - \mathcal{F}(x)).$$

For $p = 2n - j \geq n$, combine (??) and (??):

$$|S_{p,\eta}(x) - \mathcal{F}(x)| \leq 2^{-p} + 2^{1-p} e^{|\eta-1/2|} \leq 2^{-n} + 2^{1-n} e^{|\eta-1/2|}.$$

Sum against $|w_{n,j}|$ and use (??). □

For the centered shift one should not insert $\eta = 1/2$ into the deliberately coarse shift estimate. The shift difference is then exactly zero, and the sharper bound is

$$|D_n(1/2, x) - \mathcal{F}(x)| \leq V_n 2^{-n} \leq 16 2^{-n}. \quad (76)$$

This argument shows precisely why the translation parameter disappears in the limit. It is not cancelled algebraically as in the finite exact formula; rather, every fixed shift perturbs a degree- p spline by $O(2^{-p})$, and the Toeplitz rows have uniformly bounded total variation.

9.5 Exact formula versus discrete limit

The two constructions may now be compared without ambiguity.

	Exact dyadic/Appell identity	Discrete limit
Inner Thue–Morse sum	Complete block $0 \leq r < 2^k$; exact polynomial annihilation	Prefix $0 \leq r < \lfloor 2^{n+k}x + 1/2 \rfloor$; spline approximation
Outer Gaussian stencil	Extracts one coefficient exactly	Averages spline degrees $n, \dots, 2n$ with unit mass
Translation parameter	Cancels identically in the full numerator polynomial	Produces an $O(2^{-p})$ perturbation that vanishes in the limit
Passage to arbitrary x	Sum exact blocks over the binary 1-digits of x	Let spline degree tend to infinity through a regular Toeplitz transform

The formulae share the same algebraic stencil, but the reason they are valid is different. Conflating coefficient extraction with Toeplitz convergence hides the most interesting part of the story.

10 Integral-transform representations

The previous formulae read the entire function G through its Taylor coefficients. Fourier and Laplace inversion read the same product globally. They provide useful alternative representations at arbitrary, not necessarily dyadic, arguments.

10.1 Fourier–sinc inversion

Let $Y = 2X - 1$. Its density is Rvachev’s up-function on $[-1, 1]$. With the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i \xi t} dt,$$

independence gives

$$\widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^j}\right), \quad \text{sinc } u = \begin{cases} \sin u/u, & u \neq 0, \\ 1, & u = 0. \end{cases} \quad (77)$$

Since $F(x) = \mathbb{P}(Y \leq 2x - 1)$ and up is even, Fourier inversion followed by one integration gives, for $0 < x < 1$,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \frac{\sin(2\pi \xi (2x - 1))}{\xi} \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^j}\right) d\xi. \quad (78)$$

The rapid decay of the infinite sinc product makes this a genuine convergent integral, not merely a distributional formula.

The dyadic exact formulae are invisible term by term in (??); their rationality results from highly nonlocal cancellation in the integral. This contrast is useful: the Fourier form is well adapted to smoothness and global shape, while the Appell and q -forms are adapted to arithmetic at dyadic points.

10.2 Bromwich inversion of the moment product

The Laplace transform of X is

$$L_X(s) = \mathbb{E}e^{-sX} = G(-s) = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}. \quad (79)$$

For $\sigma > 0$ and $0 < x < 1$, inversion of the transform of the distribution function gives

$$F(x) = \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\sigma - iT}^{\sigma + iT} \frac{e^{sx}}{s} \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j} ds. \quad (80)$$

The same product therefore supports three complementary readings:

- coefficient extraction gives moments and exact dyadic values;
- logarithmic expansion gives cumulants and Bell-polynomial formulae;
- contour inversion gives $F(x)$ directly at an arbitrary argument.

11 Computational consequences

11.1 Exact dyadic evaluation

For exact rational computation at $x = M/2^N$, the shortest route is usually:

1. compute $d_0, \dots, d_N$ from the triangular recurrence (??);
2. form $A_m(\theta)$ by Horner evaluation or by the binomial formula (??);
3. sum only over the 1-bits in (??).

This uses $O(N^2)$ rational arithmetic in a straightforward implementation and avoids the exponentially long inner Thue–Morse sums that appear in the closed q -formula. The latter remain invaluable as structural identities and formal certificates, but are not generally the numerically cheapest way to obtain a rational value.

For inverse powers $x = 2^{-N}$, only one bit is active, so (??) collapses at once to (??). The recurrence, composition sum, determinant, Bell formula, and Gaussian-binomial formula then provide five exactly equivalent ways to compute the same rational number.

11.2 Arbitrary arguments

For a non-dyadic x , the digit–Appell series (??) is attractive because:

- zero bits cost nothing;
- each active term needs one polynomial value $A_m(\theta_m)$;
- the uniform remainder bound (??) is superexponentially small;
- no cancellation inside a 2^k -term Thue–Morse block is required.

The centered nested q -series (??) is mathematically equivalent term by term, but may be badly conditioned in ordinary floating point because large powers cancel. The raw translation freedom (??) can improve conditioning, although exact rational or ball arithmetic is preferable when evaluating the nested sums directly.

The discrete limit (??) supplies an independent numerical route and a global approximation to $\mathcal{F}$. Its error is controlled by (??); the centered shift $\eta = 1/2$ removes the shift-error term entirely, giving the sharper estimate (??), and is normally the best-conditioned canonical choice.

12 Synthesis

The apparent diversity of the formulae is reduced by separating three layers.

The moment layer. The random series (??) gives the entire moment product (??). Its coefficients are d_n , its logarithm gives the cumulants (??), and its dilation gives the recurrence (??).

The finite polynomial layer. The reduction block is the rescaled Appell polynomial

$$R_n(2^{-n}\theta) = 2^{-\binom{n}{2}} A_n(\theta)/n!.$$

The finite Thue–Morse operator removes k degrees, and the geometric Gaussian-binomial stencil removes the remaining lower-degree terms. This produces the centered and translated finite q -formulae for the same Appell polynomial.

The passage to arbitrary arguments. There are two distinct routes.

1. Follow the binary digits of x and sum exact Appell blocks. This yields (??); substitution of the finite q -identity yields (??) and (??). Dyadic rationals are exactly the terminating cases.
2. Truncate Thue–Morse power sums at the spatial point x , interpret them as splines, and apply a bounded-variation Toeplitz row. The row polynomial (??) makes the q -Pochhammer structure explicit, while (??) proves convergence quantitatively.

The central identities can be compressed into the following chain:

$$\begin{aligned}
 F(2^{-n}) &= 2^{-\binom{n}{2}} \frac{A_n(0)}{n!}, \\
 R_n(2^{-n}\theta) &= 2^{-\binom{n}{2}} \frac{A_n(\theta)}{n!}, \\
 F(x) &= \sum_{m \geq 1} b_m \varepsilon_{a_{m-1}} 2^{-\binom{m}{2}} \frac{A_m(\theta_m)}{m!}, \\
 \frac{A_m(\theta)}{m!} &= \text{one finite Gaussian-binomial/Thue–Morse block}, \\
 D_n(\eta, x) &= \sum_{j=0}^n w_{n,j} S_{2n-j,\eta}(x), \quad \sum_{j=0}^n w_{n,j} z^j = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}.
 \end{aligned} \tag{81}$$

The first four lines are exact finite algebra assembled along the binary expansion; the last line is regular summability of convergent spline approximants. Together they explain both the arithmetic exactness at dyadic rationals and the analytic convergence at arbitrary arguments.

A A compact proof of the geometric simplex sum

For positive integers $r_1, \dots, r_\ell$, put $t_h = r_h + \dots + r_\ell$. Every increasing sequence $1 \leq j_1 < \dots < j_\ell$ has a unique gap representation

$$j_i = i + g_1 + \dots + g_i, \quad g_h \in \mathbb{N} \cup \{0\}.$$

Therefore

$$\begin{aligned}
 \sum_{j_1 < \dots < j_\ell} 2^{-\sum_i j_i r_i} &= 2^{-\sum_i i r_i} \prod_{h=1}^{\ell} \sum_{g_h=0}^{\infty} 2^{-g_h t_h} \\
 &= 2^{-\sum_i i r_i} \prod_{h=1}^{\ell} \frac{1}{1 - 2^{-t_h}}.
 \end{aligned}$$

But $\sum_h t_h = \sum_i i r_i$, so

$$2^{-\sum_i i r_i} \prod_h \frac{2^{t_h}}{2^{t_h} - 1} = \prod_h \frac{1}{2^{t_h} - 1},$$

which is (??).

B First moments and inverse-dyadic values

The recurrence (??) gives:

n	$d_n = \mathbb{E}X^n$	$F(2^{-n}) = 2^{-\binom{n}{2}} d_n / n!$
0	1	1
1	1/2	1/2
2	5/18	5/72
3	1/6	1/288
4	143/1350	143/2073600

The $n = 0$ row uses the standard convention $2^0 = 1$ and $0! = 1$.

C Map to the formal development

The principal formal ingredients used here are organized in the following source files in the `Analysis/FabiusFunction` directory of the `ProveIt` repository:

File	Mathematical role
<code>HalfQBinomial.lean</code>	finite q -Pochhammer identities, Gaussian binomial coefficients at $q = 1/2$, and the finite q -binomial annihilator
<code>FabiusQBinomialFormula.lean</code>	centered exact formulae for inverse and general dyadic arguments
<code>FabiusRawQBinomialFormula.lean</code>	raw-coordinate formula with arbitrary common translation
<code>FabiusQBinomialTaylor.lean</code>	polynomial identity equating the q -binomial block with the Taylor reduction polynomial
<code>TaylorReduction.lean</code>	exact reduction polynomial R_n and dyadic-shell Taylor identities
<code>FabiusBinaryReductionSeries.lean</code>	finite telescoping identity, residual estimate, and global binary series
<code>FabiusGlobalQBinomialSeries.lean</code>	termwise substitution of the finite q -block into the binary series
<code>FabiusUniformSpline.lean</code>	centered finite splines and uniform convergence
<code>FabiusComplexShiftSpline.lean</code>	complex translation of the spline and quantitative shift stability
<code>FabiusDiscreteLimitToeplitz.lean</code>	Toeplitz weights, row mass, and total-variation bound
<code>FabiusDiscreteLimitIntegration.lean</code>	exact reindexing of the discrete formula and convergence to the global Fabius extension

References

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